

§ 11.4 The Comparison Tests(비교판정법)

Ex.4 까지

■ The Direct Comparison Test (직접비교판정법)

Let's consider the two series

$$\sum_{n=1}^{\infty} \frac{1}{2^n + 1} \quad \text{and} \quad \sum_{n=1}^{\infty} \frac{1}{2^n}$$

The Direct Comparison Test Suppose that $\sum a_n$ and $\sum b_n$ are series with positive terms.

- (i) If $\sum b_n$ is convergent and $a_n \leq b_n$ for all n , then $\sum a_n$ is also convergent.
- (ii) If $\sum b_n$ is divergent and $a_n \geq b_n$ for all n , then $\sum a_n$ is also divergent.

In using the Direct Comparison Test we must, of course, have some known series $\sum b_n$ for the purpose of comparison. Most of the time we use one of these series:

- A p -series [$\sum 1/n^p$ converges if $p > 1$ and diverges if $p \leq 1$]
- A geometric series [$\sum ar^{n-1}$ converges if $|r| < 1$ and diverges if $|r| \geq 1$]

EXAMPLE 1 Determine whether the series $\sum_{n=1}^{\infty} \frac{5}{2n^2 + 4n + 3}$ converges or diverges.

Solution:

EXAMPLE 2 Test the series $\sum_{k=1}^{\infty} \frac{\ln k}{k}$ for convergence or divergence.

Solution:

■ Limit Comparison Test (극한비교판정법)

Consider $\sum_{n=1}^{\infty} \frac{1}{2^n - 1}$

The Limit Comparison Test Suppose that $\sum a_n$ and $\sum b_n$ are series with positive terms. If

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c$$

where c is a finite number and $c > 0$, then either both series converge or both diverge.

EXAMPLE 3 Test the series $\sum_{n=1}^{\infty} \frac{1}{2^n - 1}$ for convergence or divergence.

Solution:

EXAMPLE 4 Determine whether the series $\sum_{n=1}^{\infty} \frac{2n^2 + 3n}{\sqrt{5 + n^5}}$ converges or diverges.

Solution:

